Principal Component Analysis

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Introduction

Principal component analysis (PCA) takes a data matrix with many correlated variables and re-expresses it in a smaller number of uncorrelated components that capture most of the variance. It is used for exploratory visualisation, for compression, as a preprocessing step for regression and classification, and as a quality-control tool in high-dimensional experiments such as RNA-seq. It is not a model for the data; it is a change of basis.

Prerequisites

The reader should be familiar with vectors, variance, covariance, and correlation. Some understanding of eigenvalues and eigenvectors helps but is not essential.

Theory

Given a data matrix $\mathbf{X}$ with n rows (observations) and p columns (variables), PCA finds orthogonal directions in the p-dimensional space of variables that successively maximise the projected variance. The first principal component (PC1) is the direction of greatest variance; PC2 is the direction of greatest variance orthogonal to PC1; and so on up to $\min(n-1, p)$.

Algebraically, if $\mathbf{X}$ is centred (each column has mean zero), PCA is the eigendecomposition of the covariance matrix $\mathbf{S} = \frac{1}{n-1}\mathbf{X}^\top\mathbf{X}$:

$$\mathbf{S} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top,$$

where the columns of $\mathbf{V}$ are the principal-component directions (loadings), and $\boldsymbol{\Lambda}$ is the diagonal matrix of variances along each direction (eigenvalues). The scores – coordinates of the observations in the new basis – are $\mathbf{Z} = \mathbf{X}\mathbf{V}$.

A practical issue: if the variables are on different scales, the variance-maximisation criterion is dominated by whichever variable happens to have the biggest numeric range. For heterogeneous data, PCA should be performed on the correlation matrix instead – equivalent to standardising each variable to unit variance before the decomposition.

Assumptions

PCA has no formal probabilistic assumptions, but its interpretation rests on:

1. The relationships of interest are linear in the variables.
2. Variance is a reasonable proxy for “importance”.
3. The first few components contain most of the useful signal.

The first is violated when relationships are non-linear; alternative methods (kernel PCA, UMAP) may be needed. The second is violated when small-variance components encode the scientific signal (for example, subtle gene regulators in a bulk RNA-seq study).

R Implementation

```r
library(FactoMineR)
library(factoextra)

data(iris)
X <- iris[, 1:4]

pca <- PCA(X, scale.unit = TRUE, graph = FALSE)

fviz_eig(pca, addlabels = TRUE)
fviz_pca_biplot(pca,
                habillage = iris$Species,
                palette = "Set2",
                addEllipses = TRUE,
                label = "var")

summary(pca)
```

`PCA()` from `FactoMineR` handles centring and scaling by default. `fviz_eig()` draws the scree plot; `fviz_pca_biplot()` draws the biplot with observations coloured by species and variable-loading arrows overlaid. `summary()` produces a textual summary including variance explained, variable loadings, and contribution percentages.

Output & Results

For the `iris` data, the first component captures approximately 73% of the variance, the second another 23%; together the first two components retain 96% of the information. The biplot shows that the three species separate cleanly along PC1, which is heavily loaded on petal length, petal width, and sepal length.

Interpretation

In a manuscript, a PCA is usually reported in two pieces: the proportion of variance explained by the first k components, and the biological or physical interpretation of each component's loadings. "PC1 (73% of variance) is a size component, loading positively on all length measurements; PC2 (23%) contrasts sepal width against the other measurements and separates *I. setosa* from the other two species."

Practical Tips

- Always standardise (`scale.unit = TRUE`) unless all variables share the same units and you have a reason to preserve absolute-scale differences.
- Inspect the scree plot and keep components up to an elbow, or enough for 80-90% cumulative variance, or until an interpretable pattern emerges. There is no universal rule.
- The sign of each component is arbitrary (eigenvectors are unique up to sign), so "PC1 goes up" and "PC1 goes down" are equivalent representations; do not over-interpret the direction.
- For high-dimensional sparse data (e.g., scRNA-seq), use randomised SVD (`irlba`, `rsvd`) or the PCA implementations in `Seurat` or `scran` rather than dense eigendecomposition.
- PCA is sensitive to outliers; robust variants (ROBPCA, projection pursuit PCA) are available in the `rrcov` package.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**

- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE